

Ambo University
College of Business and Economics
Department of Economics

Model Exam for Economics 4th Year Regular 2015 (2022/2023)

General Guideline

- ⌚ All questions are multiple-choice type
- ⌚ Weight of the exam is 100%
- ⌚ Time allotted to complete is 100 minute

Direction: Choose appropriate answer those alternative

Basic Micro

1. Cardinal utility theory assumes that consumers can
 - A. rank baskets of goods as to their preference.
 - B. determine the number of utils that can be derived from consuming all goods.
 - C. determine the marginal rate of substitution between goods.
 - D. avoid the law of diminishing marginal utility.
2. If for a rational consumer the MU_x is twice as much as the MU_y then the consumer reaches equilibrium when,
 - A. The price of good Y is twice price of X.
 - B. The price of good X is half of price of Y.
 - C. The price of good X is twice price of Y.
 - D. Price of good X is 2 birr.
3. When output is zero,
 - A. TVC is zero.
 - B. TC is not necessarily zero.
 - C. TC equals TFC.
 - D. All
4. The main difference between perfect competition & monopolistic competition is:
 - A. The number of sellers in the market.
 - B. The ease of exit from the market.
 - C. The difference in the firm's profits in the long run.
 - D. The degree of product differentiation.
5. Excess capacity in monopolistically competitive firm results because in equilibrium
 - A. Each firm's output is too great to minimize average cost.
 - B. Each firm's output level is too small to minimize average cost.
 - C. Firms make positive profit.
 - D. All are the above.
6. Consider Cournot's model of oligopoly with zero cost of production in the long run,
 - A. Each firm will produce one half of the market.
 - B. Each firm will produce one third of the market.
 - C. Both firms will produce one half of the market.
 - D. It is indeterminate
7. Which one of the following is true about Stackelberg and Cournot duopoly models?
 - A. Firms will react to the price of the rival firm.
 - B. Firms will maximize profits simultaneously in Stackelberg, sequentially in Cournot.

- C. Firms will maximize profits simultaneously in Cournot, sequentially in Stackelberg.
D. Both models are collusive.
8. The long run is a period of time in which:
- A. the firm may want to build a bigger plant, but cannot do so
B. the quantities of all inputs can be varied.
C. economic efficiency is achieved
D. All of the above
9. The '*law of diminishing marginal returns*' refers to the general tendency for ____ to eventually diminish as more of the variable input is employed, given the quantity of fixed inputs.
- A. Total product of Labour
B. Average Product of Labour
C. Marginal Product of Labour
D. None of the above
10. When the marginal product of labour is greater than the average product of labour:
- A. the average product of labour is increasing
B. the firm is experiencing diminishing returns
C. the marginal product of labour is increasing
D. the total product curve is negatively sloped
11. The intersection of marginal product curve and average product curve characterizes the point of
- A. Maximum total product
B. Maximum marginal product
C. Maximum average product
D. Zero marginal product.
12. Assume the demand curve of the monopolist is given by $Q = 4 - P$ then the MR will be
- A. $MR = 4 - P$
B. $MR = 4 - Q$
C. $MR = 4 - 2P$
D. $MR = 4 - 2Q$
13. Monopoly power of a firm results from the ability to
- A. Set price equal to Marginal cost.
B. Equate MR to MC.
C. Set Price above the MC
D. Set price below Marginal cost

14. Assume a coffee seller classifies her customers of coffee as students, administrative workers and teachers and then charges each group 1.5 birr, 2.5 birr and 3 birr respectively; then which type of price discrimination she is practicing?

- A. First Degree
- B. Second Degree
- C. Third Degree
- D. The act is not price discrimination

15. The optimal (best) choice of strategy for each player no matter what the other player does is

- A. Dominant strategy
- B. Nash equilibrium
- C. Prisoners' dilemma
- D. All of the above

16. According to Pareto, the standard equilibrium conditions for the consumer side of economy is where

- A. Marginal rate of substitution equal to the price ratio.

B.
$$MRS_{12}^A = \left(\frac{MU_1}{MU_2} \right)^A = \frac{P_1}{P_2} = MRS_{12}^B = \left(\frac{MU_1}{MU_2} \right)^B$$

- C. The highest Indifference curve of the consumers is tangent to the budget (price) line.
- D. All of the above.

17. Which of the following statements is true?

- A. A game that does not have a Nash equilibrium under pure strategies will not have a Nash equilibrium under mixed strategies.
- B. Dominant strategy equilibrium exists for a game in which only one of two firms has a dominant strategy.
- C. The difference between pure and mixed strategies lies on the number of strategies available to the players.
- D. In a situation where Nash equilibrium does not yield a Pareto efficient outcome, firms face the problem of prisoners' dilemma.

18. The prisoners' dilemma game yields an equilibrium in which:

- A. both criminals choose not to confess, but would be better off by confessing.
- B. both criminals choose to confess, but would be better off by not confessing.
- C. both criminals could get the lightest sentence possible if they both confess.
- D. both criminals would receive the maximum sentence if they both confess.

19. Tangible benefits in Economic Analysis includes all **except**;

- A) Increased benefits through Externality
- B) Quality improvement
- C) Changes in time of sale
- D) Changes in location of sale

20. A project is said to be sustainable if:

- A) Its products or services meet the needs of the local community
- B) It generates some income and becomes self-reliant financially
- C) It protects the environment
- D) All

21. The advantages with Bottom-up approach **includes**

- A) Capacity is built within the community
 - B) To obtain quick results
 - C) There is increased use of local resources including labor
 - D) A & C**
22. Identify the one which is not the reason for undertaking project from the following;
- A. powerful and efficient means to achieve development (growth)
 - B. are mechanisms for improving income distribution
 - C. mechanisms to solve immediate problems
 - D. all**
23. The project cycle under which pre-feasibility and feasibility study is mandatory is;
- A. UNIDO project cycle
 - B. Baum project cycle**
 - C. New project cycle
 - D. None
24. Appraisal phase is
- A) A phase of implementation-activity phase
 - B) A phase of evaluation on impact of the project
 - C) A phase of **decision making whether** to accept or not to accept the project
 - D) All

Basic Macro

25. International trade is based on the idea that:
- A. Exports should exceed imports
 - B. Imports should exceed exports
 - C. Resources are more mobile internationally than are goods
 - D. Resources are less mobile internationally than are goods**
26. A primary reason why nations conduct international trade is because:
- A. Some nations prefer to produce one thing while others produce another
 - B. Resources are not equally distributed to all trading nations**
 - C. Trade enhances opportunities to accumulate profits
 - D. Interest rates are not identical in all trading nations
27. International trade tends to cause welfare losses to at least some groups in a country
- A. The less mobile the country's resources**
 - B. The more mobile the country's resources
 - C. The lower the country's initial living standard
 - D. The higher the country's initial living standard

Use the information in the table below to answer the next three questions.

<u>Country</u>	<u>Tons of steel</u>	<u>DVDs</u>
South Korea	80	40
Japan	20	20

28. The opportunity cost of one DVD in Japan is:

- A. One ton of steel
 - B. Two tons of steel
 - C. Three tons of steel
 - D. Four tons of steel
29. According to the principle of comparative advantage:
- A. South Korea should export steel
 - B. South Korea should export steel and DVDs
 - C. Japan should export steel
 - D. Japan should export steel and DVDs
30. With international trade, what would be the maximum amount of steel that South Korea would be willing to export to Japan in exchange for each DVD?
- A. One-half ton of steel
 - B. One ton of steel
 - C. Two tons of steel
 - D. Two and one-half tons of steel
31. Wassily Leontief used an input-output table in order to test the
- A. Ricardian theory of comparative advantage
 - B. Heckscher-Ohlin theory of comparative advantage
 - C. Linder theory of overlapping demand
 - D. All of the above
32. A tax of 15 percent per imported item would be an example of a(n):
- A. Ad valorem tariff
 - B. Specific tariff
 - C. Effective tariff
 - D. Compound tariff
33. Which of the following is not the similarity between Tariff and Quota?
- A. both raise the import price and reduce the import quantity
 - B. Both bring revenue to the government.
 - C. Both cut imports to a certain quantity
 - D. They produce similar effects on consumption, production, etc.
34. Which industrialization policy has developing countries used which places emphasis on the comparative advantage principle as a guide to resource allocation:
- A. export promotion
 - B. import substitution
 - C. international commodity agreements
 - D. multilateral contracts
35. Export-led growth strategies tend to emphasize
- A. resource allocation based on the principle of absolute advantage
 - B. resource allocation based on the principle of comparative advantage
 - C. trade protection for import-competing firms
 - D. trade protection for exporting-competing firms
36. Which of the following statements is incorrect about the foreign exchange market?
- A) All countries must have their own unit of account to participate in the foreign exchange market.
 - B) The two purchases in the foreign exchange market are foreign currency is bought and facilitation of international transaction.
 - C) Foreign exchange market has no centralized meeting place and no formal requirements for participation.

- D) Retail market is the mutual adjustment process between the banks regarding the demand and supply of foreign exchange.

37.Regarding exchange rate determination, which of the following is wrongly matched?

- A) Foreign exchange arbitrage refers to the purchasing of a foreign currency where its price is low and selling it where the price is high.
- B) Depreciation is a rise in the exchange rate reduction in the value of the domestic currency in relation to the foreign currency.
- C) Appreciation is a fall in the exchange rate or an increase in the value of the domestic currency in relation to the foreign currency.
- D) Appreciation is a rise in the exchange rate or a decline in the value of the domestic currency in relation to the foreign currency.

38.Concerning money, interest rate and exchange rate which of the following is correct?

- A) At market equilibrium of money, if there is initially an excess money supply, the interest rate falls, and if there is initially an excess demand for money, interest rates will rises.
- B) In money market equilibrium, if there is initially shortage of money supply, the interest rate increases, and if there is initially a shortage of demand for money, interest rate will rises.
- C) An increase in a country's money supply causes its currency to appreciate in the foreign exchange market, while a reduction in the money supply causes its currency to depreciate.
- D) An increase in a country's money supply causes its currency to depreciate in the foreign exchange market, while a reduction in the money supply causes its currency to appreciate.

39.Which of the following is incorrect regarding the purchasing power parity of certain country?

- A) It asserts that all countries' price levels are equal when measured in terms of the same currency in a given country's economy.
- B) It provides that prices and exchange rates change in a way that preserves the ratio of each currency's domestic and foreign purchasing powers.
- C) The notion of relative purchasing power parity comes in handy when we have to rely on government price level statistics to evaluate purchasing power parity.

D) The perception of absolute purchasing power parity comes in handy when we have not relied on government price level figures to evaluate purchasing power parity.

40. Of the following questions, which one is/are not correct statement regarding balance of payments?

- A) It reveals how many goods and services the country has been exporting and importing during the particular calendar.
- B) It shows whether the country has been borrowing from or lending money to the rest of the world during the fiscal year.
- C) The balance of payments is a statistical record of all the economic transactions between the reporting country and the rest of the world during a year.

D) Central monetary authority ignores addition to or reduction from its reserves of foreign currency report in the statistics.

41. If government stimulates domestic spending by increasing government purchases or by cutting taxes, what a type of policy is implemented?

- A) Expansionary monetary policy
- C) Expansionary fiscal policy
- B) Contractionary fiscal policy
- D) Contractionary monetary policy

42. What a kind of balance of payment approach provides an analysis of what happens to the current account balance when the country devaluates its currency?

- A) Absorption approach of balance of payment
- B) Elasticity approach of balance of payment
- C) Monetary approach of balance of payment
- D) Income elasticity approach of balance of payment

43. Economic development differs from economic growth in the following ways *except one*

- A. The economic growth is narrower than the Economic development
- B. The economic growth contain the Economic development
- C. The Economic development show welfare and standard of living
- D. Economic development consist of the economic growth

44. Which one of the following is among the common characteristics of the third world countries?

- A. High levels of living
- C. High rates of population growth
- B. Low levels of productivity
- D. High levels of unemployment

45. Which one of the following is **not true about diverse structure of the third world countries**

- A. The country with larger physical size will always have higher per capita income.
- B. Africans have a shorter history of political independence and less shared colonial heritage
- C. Landlocked and tropical countries seem to do worse than temperate and Island countries.

46. Suppose that Ethiopia planned to register equilibrium growth rate of 20%. Assume that the economy Ethiopia is currently operating at a capacity production level of **500** per year and the capital stock is **1000**. How much percent of her national income should be saved in order to achieve the targeted growth if the country implemented Harrod-Domar model?

A. 60% B. 30% C. 20% D. **40%**

A. Pre-conditions for the take-off
B. Take-off
C. Drive to maturity
D. Age of high mass consumption.

A. Big push theory
B. Balanced growth theory
C. Unbalanced growth theory
D. Harrod Domar Growth model

A. Type of government
B. Capital accumulation
C. Social structure
D. Work culture and discipline

- A. The closer Lorenz curve to line of perfect equality, the higher inequality of income.
- B. The lower the Kuznet ratio, the lower inequality will be.
- C. Shift from a traditional to a modern economy, indicate the rise in inequality.
- D. Absolute poverty is a serious worry of advanced countries.

A. Population distribution
B. Subordination of women
C. World resource depletion and environmental destruction
D. An undesirable phenomenon of Population growth

- A. A decrease in quality of education
- B. The **match** between knowledge and skills acquired at schools and market demand
- C. Wage difference for worker with higher and lower education and skills.
- D. The mismatch between knowledge and skills acquired at schools and market demand

- A. Ensuring an appropriate rural-urban economic imbalances
- B. Contraction of small-scale capital intensive industries
- C. Contraction of small-scale labor intensive industries
- D. Centralizing authority to cities and neighborhoods

54.One of the following doesn't describe the benefit of agriculture to economic development

- a. Possibility to transfer surplus manpower to other sectors
 - b. Unlike other sectors, relief from shortage of capital
 - c. Assist to remove economic depression
 - d. **Withdraw** funds from capital formation
- 55.** Which one of the following is the role of foreign trade in economic development
- a. Deteriorate quality
 - b. **Discourage** monopoly power
 - c. Result in price instability
 - d. Threatening world peace
- 56.** One is the supply side factors that limit expansion of primary-commodity export
- a. Low income elasticity
 - b. Development of synthetic substitutes
 - c. Low population growth rates developed countries
 - d. The structural **rigidity** of many rural production systems
- 57.** The desired number of children are expected to vary
- A. Inversely with the price of children
 - B. Inversely with the strength of tastes for other goods
 - C. Positively with the level of household income
 - D. **Inversely** with price of all other goods
- 58.** Which one of the following is true regarding the influence of foreign debt on economy
- A. Using up available debt capacity today will mean reduction of future flexibility
 - B. If foreign debt is used for capital investment, it **will depress incentives** for investments.
 - C. If foreign debt is used on nonproductive projects, it will depress savings and create inflation.
 - D. Foreign debt has tendency to increase or decrease domestic consumption
- 59.** The term "Liquidity" refers to
- A. The process of fixing appropriate interest rate by the central bank.
 - B. The process of supplying adequate loanable funds by the commercial banks.
 - C. The ease with which an asset can be exchanged for other assets.
 - D. The process of determining appropriate money supply by the Central bank.
- 60.** Which one of the following is part of Chicago approach of money measurement, but not part of conventional approach?
- A. Currency held in commercial banks
 - B. Currency held in circulation
 - C. **Time deposits**
 - D. Demand deposits
- 61.** Which one of the following is the correct expression?
- A. $M^s = (1/m)MB$
 - B. $M^s = \frac{c+e+r}{1+c} MB$
 - C. $MB = (c+e+r) D$
 - D. $MB = \frac{1+c}{c+e+r} M^s$
- 62.** Identify the incorrect statement about money multiplier?
- A. It tells us what multiple of the monetary supply is transformed into the monetary base.
 - B. It reflects the effect on the money supply of other factors besides the monetary base.
 - C. **It tells us what multiple of the monetary base is transformed into the money supply.**
 - D. It is negatively related to the required reserve ratio.
- 63.** The **primary reason** for money supply to increase when discount rate decreases is because of the fact that

- A. Less discount rate implies that more MB to support Deposits (D) and Currency (C).
- B. Less discount rate reflects less multiple deposits system expansion.
- C. Less discount rate means that less monetary base to increase money supply.
- D. Less discount rate implies that more required reserves banks are required to put in central banks.

64. The most important, effective and commonly used policy tool(s) that the Central Banks use to control the money supply is (are)

- A. Open Market Operation (OMO)
- B. Discount Rate (DR)
- C. Required Reserve Ratio (r)
- D. A and B

65. According to Friedman the lower the returns of other assets (such as bonds, equities and goods) relative to the expected return on money,

- A. The lower the quantity of money demanded.
- B. The higher the quantity of money demanded.
- C. The lower the quantity of money supplied.
- D. Indeterminate demand for money.

66. Liquidity preference theory of Keynes mainly deals with

- A. How average price is determined in an economy
- B. Why people prefer to hold certain amount of real balances
- C. How velocity of money circulation is calculated
- D. How demand for money is a function of other assets that are substitutes for money.

67. The money multiplier is the ratio of

- A. bank deposits divided by bank reserves
- B. bank deposits divided by high-powered money
- C. money supply divided by high-powered money
- D. the currency divided by bank reserves

68. Any position that is to the right of (and below) the LM-curve indicates that there is

- A. excess demand for goods and services
- B. excess supply of goods and services
- C. excess demand for money
- D. excess supply of Money

69. The AD-curve will shift to the right if

- A. government transfer payments are increased
- B. real money balances increase due to a decrease in the price level
- C. autonomous saving increases
- D. the central bank restricts nominal money supply

70. If domestic saving is less than domestic investment, then net exports are _____ and net capital outflows are _____.

- A. positive; positive.
- B. positive; negative.
- C. negative; positive.
- D. negative; negative.

71. Unemployment caused by the time it takes workers to search for a job is called _____ unemployment.

A. frictional

B. structural

C. insider

D. efficiency

72. An increase in the price of goods bought by firms and the government will show up in:

A. the GDP deflator but not in the CPI.

B. the CPI but not in the GDP deflator.

C. neither the CPI nor the GDP deflator.

D. both the CPI and the GDP deflator.

73. In the classical model with fixed income, if households want to save more than firms want to invest, then:

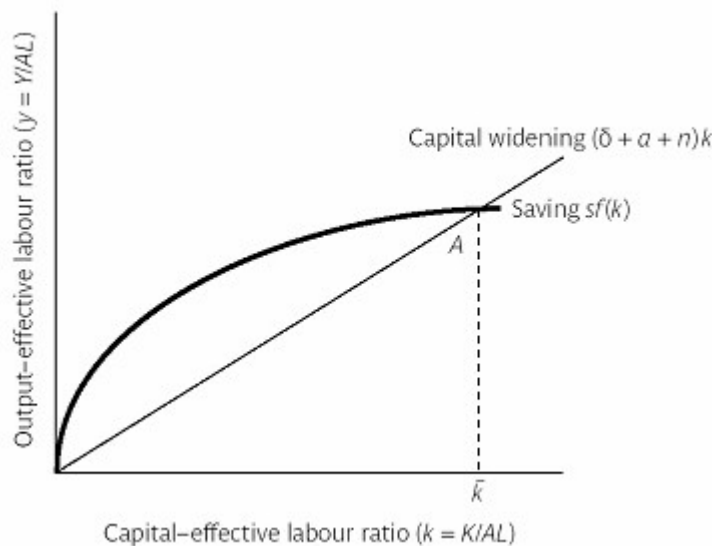
A. output increases.

B. output falls.

C. the interest rate rises.

D. the interest rate falls.

74. The figure below represents an economy with a population growing at rate n with an unchanging labour force participation rate. The rate of technological progress is a and the rate of depreciation is δ .



In the steady state of this model, output per unit labour...

A. will grow at the rate a .

B. will grow at the rate $a+n$.

C. will grow at the rate $a - n$.

D. will grow at the rate $\delta + a + n$.

75. Bank reserves equal:

A. gold kept in bank vaults.

B. gold kept at the central bank.

C. currency plus demand deposits.

D. deposits that banks have received but have not lent out.

76. Starting from a long-run equilibrium, if a drought pushes up food prices throughout the economy, the Fed could move the economy more rapidly back to full output by:

- A. decreasing the money supply, but at a cost of permanently lower prices.
- B. increasing the money supply, which would restore the original price level.
- C. decreasing the money supply, which would restore the original price level.
- D. increasing the money supply, but at a cost of permanently higher prices.

77. The Phillips curve shows a _____ relationship between inflation and unemployment, and the short-run aggregate supply curve shows a _____ relationship between the price level and output.

- A. positive; positive
- B. positive; negative
- C. negative; positive
- D. negative; negative

- 78.** Stabilization policy refers to policy actions aimed at:
- A. reducing the severity of short-run economic fluctuations.
 - B. equalizing incomes of households in the economy.
 - C. maintaining constant shares of output going to labor and capital.
 - D. preventing increases in the poverty rate.
- 79.** If policymakers announce in advance how policy will respond to various situations and commit themselves to following through on this announcement, this is:
- A. policy by rule.
 - B. monetary policy.
 - C. policy by discretion.
 - D. time inconsistent policy.
- 80.** Wage rigidity:
- A. prevents labor demand and labor supply from reaching the equilibrium level.
 - B. increases the rate of job finding.
 - C. forces labor demand to equal labor supply.
 - D. is caused by sectorial shifts.
- 81.** Because monetary and fiscal lags are long and variable:
- A. stronger policies must be used.
 - B. successful stabilization policy is completely impossible.
 - C. attempts to stabilize the economy are often destabilizing.
 - D. policy must be completely passive.
- 82.** The aggregate production function for the Solow growth model assumes (A) _____ returns to scale and (B) _____ marginal productivity of labour and capital.
- A. increasing: diminishing
 - B. constant: diminishing
 - C. decreasing: constant
 - D. constant: increasing

Quantitative

- 83.** The first four derivatives of the function $f(x) = -x^6$ is
- A. $-360x^2$
 - B. $-120x^{-6}$
 - C. $720x$
 - D. $-720x$
- 84.** $\int_0^4 (3x \, dx)$ equals
- A. 21
 - B. 24
 - C. 48
 - D. 12
- Given the optimization of $Z = 60x + 34Y - 4XY - 6X^2 - 3Y^2 + 5$ answer question [86-89]
- 85.** The critical values are;
- A. (3,4)
 - B. (4,3)
 - C. (6,4)
 - D. (4,6)
- 86.** The second order partial derivatives of Z_{xx} equals;
- A. -4
 - B. 60
 - C. -12
 - D. 4
- 87.** Cross partials of Z_{xy} equals;
- A. -6
 - B. -4
 - C. -12
 - D. 34

88. Based on all the above result, the function reach at;

- A. Inflection point B. Saddle point C. Maximum point D. Minimum point

89. Given the following constraints function, which of the following x_1 and x_2 maximize the given constraint function

$$\max y = 10x_1 - x_1^2 + 180x_2 - x_2^2, \text{ subject to } x_1 + x_2 \leq 80, x_1, x_2 \geq 0,$$

- A. (0,40) B. (80,0) C. (0,80) D. (40,80)

90. What is general solution for this Bernoulli dynamic equation $\frac{dy}{dt} + ty = te^{-t^2} y^{-3}$, where $n = -3$

- A. $y^4 = 2e^{-t^2} + ce^{-2t^2}$ C. $y = Ae^{-vt}$
 B. $y_t = zt + y(0)$ D. $y(t) = A^{\frac{1}{1-a}} \left(\frac{s}{(\delta+n)} \right)^{\frac{a}{1-a}}$

91. Which one of the following is not correct considering the definition of economic and econometric model?

- A. Econometric methods are generally developed for the analysis of experimental data.
 B. An economic model is a set of assumptions that describes the behavior of an economy.
 C. An econometric model consists of A set of behavioral equation derived from the economic model.
 D. Econometrics can be defined as 'the application of mathematics and statistical methods to the analysis of economic data.'

92. Suppose Gada Company wants to know the relationship between sales and advertising.

Based on the following data what is change of sale due to one extra change of advertising?

(Hint use simple linear regression model)

$\Sigma XY = 28202$	$\Sigma X^2 = 9253$	$\Sigma Y = 589$
$N = 4$	$\Sigma X = 189$	

- A. 0.75 C. 0.65
 B. 1.15 D. 2.15

93. Based on the above data how much Gada company is going to sale when there is no advertising?

A. 92.9

C. 153.25

B. 47.25

D. 4.724.

94. Improper use of dummy variables highly has impact on.

A. Multicollinearity

C. Simultaneity

B. Autocorrelation

D. Endogeneity

95. Which one of the following assumptions of CLR violation is happen because of high variability among variables

A. Heteroscedasticity

C. Non normality

B. Multicollinearity

D. Autocorrelation

96. What is the main objective of researcher when dependent variable is dummy?

A. To calculate the mean change of Y due to X

B. To calculate the mean change of X due to Y

C. To calculate the probability, change of Y due to X

D. To calculate the probability of X due to Y

97. Suppose Economic model $Y_i = \alpha_1 + \alpha_2 D_{2i} + \alpha_3 D_{3i} + \beta X_i + u_i$ what is Mean salary for black male professor?

Where Y_i = annual salary

X_i = years of teaching experience

$D_2 = 1$ if male
=0 otherwise

$D_3 = 1$ if white
=0 otherwise

A. $(\alpha_1 + \alpha_2 + \alpha_3) + \beta X$

C. $(\alpha_1 + \alpha_2) + \beta X_i$

B. $(\alpha_1 + \alpha_3) + \beta X_i$

D. $(\alpha_1 + \alpha_3) + \beta X_i$

98. The two-way causation in a relationship leads to violation of the important assumption of linear regression model that referred to as

A. Simultaneity bias of OLS estimators

C. Reduced form bias.

B. System of simultaneous equation

D. Structural model bias

99. Assume Ambo university fourth year students conducting research and the results of the regressions is $GDP = 28.2054 - 0.00136GDP_{t-1}$

$t = (1.1576) (-0.2191)$

$R^2 = 0.00056$

$d = 1.35$

1. GDP is stationary.

3. GDP is stochastic.

2. GDP not stationary

4. GDP not stochastic

100. A test of stationarity (or non-stationarity) that has become widely popular over the past several years is;

A. Unit root test.

C. Goldfery test

B. Bruch pagan test

D. Deckey fuler test